

Complex Analysis Qualifying Exam Review

Fall 2026

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Chapter 1

Fall 2025

Problem 1.1 (Theorem 5.2, Stein, Chapter 2). Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \rightarrow f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \rightarrow \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Proof. This is an immediate consequence of Morera's theorem. Take an arbitrary triangle T contained in $\mathbb{D}$, we know for any n ,

$$\int_T f_n(z) dz = 0$$

by Goursat's theorem. By assumption, $f_n \rightarrow f$ uniformly, thus

$$\int_T f(z) dz = \lim_{n \rightarrow \infty} \int_T f_n(z) dz = 0$$

By Morera's theorem, f is holomorphic in $\mathbb{D}$. □

Problem 1.2. Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

Proof. Use residue theorem for $z_1 = e^{\frac{\pi i}{4}}$, $z_2 = e^{\frac{3\pi i}{4}}$ and the semicircle along the real axis, we know

$$2I + \int_{\gamma_R} \frac{z^2}{z^4 + 1} dz = 2\pi i (\text{res}_{z_1} f + \text{res}_{z_2} f)$$

We first show the second term goes to zero.

$$\begin{aligned} \left| \int_{\gamma_R} \frac{z^2}{z^4 + 1} dz \right| &= \left| \int_0^\pi \frac{R^2 e^{2i\theta}}{R^4 e^{4i\theta} + 1} R i e^{i\theta} d\theta \right| \\ &\leq \frac{C}{R} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Now using the residue formula, we compute

$$\text{res}_{z_1} f = \frac{1}{4e^{\pi i/4}}, \quad \text{res}_{z_2} f = \frac{1}{4e^{3\pi i/4}}$$

Thus,

$$I = \frac{\pi}{2\sqrt{2}}$$

□

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

Proof. If we can show $1/f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic, then it suffices to show $|1/f(z)| \leq 1$. Fix any $0 < r < 1$, by MMP, for $|z| < r$,

$$|1/f(z)| \leq r^{-1/2}$$

Thus letting $r \rightarrow 1-$, we get $|f(z)| \leq 1$. Now it remains to show $1/f$ is well-defined on $\mathbb{D}$, i.e., f never vanishes. Since $|f(z)| \geq |z|^{1/2}$, the only possible zero would be $z = 0$. Suppose $f(0) = 0$, then $f(z) = z^n g(z)$ for some $n \geq 1$ and $g(0) \neq 0$, where g is holomorphic. Then

$$|z|^m |g(z)| \geq |z|^{1/2} \Rightarrow |z|^{1/2-m} \leq C$$

which is a contradiction if we let $|z| \rightarrow 0$. □

Problem 1.4. 8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Proof. We will use Rouché's theorem. First we count the number of zeros of $f(z) = z^7 - 7z^4 + z^3 - 3z - 1$ inside of $|z| < 2$. Let

$$g_1(z) = z^7, \quad g_2(z) = -7z^4 + z^3 - 3z - 1$$

Then on $|z| = 2$,

$$|g_2(z)| < |g_1(z)|$$

Thus f has 7 zeros inside of $|z| < 2$. Next we count the number of zeros inside $|z| < 1$. Let

$$h_1 = -7z^4, \quad h_2 = z^7 + z^3 - 3z - 1$$

On $|z| = 1$, we see

$$|h_2(z)| < |h_1(z)|$$

Thus f has 4 zeros inside $|z| < 1$. Then f has 3 zeros in $\{z : 1 \leq |z| < 2\}$. It suffices to consider the zeros on $|z| = 1$. We notice that

$$\begin{aligned} |f(z)| &\geq |7z^4| - |z^7 + z^3 - 3z - 1| \\ &\geq 7 - (|z^7| + |z^3| + 3|z| + 1) \\ &\geq 2 \end{aligned}$$

thus f has 3 zeros in the annulus $\{z : 1 < |z| < 2\}$. □

Chapter 2

Cauchy's integral formula

Proposition 2.1 (Cauchy's theorem). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U , if γ is a closed smooth curve whose interior is contained in U , then

$$\int_{\gamma} f(z) dz = 0$$

Proposition 2.2 (Cauchy's integral formula). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then for any $z \in D_r(z_0)$,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw$$

where $\gamma = z_0 + re^{i\theta}$, $\theta \in [0, 2\pi]$. Moreover,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{n+1}} dw$$

Proposition 2.3 (Cauchy's inequality). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then

$$|f^{(n)}(z_0)| \leq \frac{n! \sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.4 (Liouville's theorem). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a bounded entire function, then f must be constant.

Proposition 2.5 (Holomorphic and analytic). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then f has a power series expansion at z_0 :

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D_r(z_0)$, where

$$a_n = \frac{f^{(n)}(z_0)}{n!}$$

This combined with the previous theorem gives a bound on the coefficients:

$$|a_n| \leq \frac{\sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.6 (Morera's theorem). Suppose f is a continuous function in an open set U , if for any triangle T contained in D ,

$$\int_T f(z) dz = 0$$

then f is holomorphic.

Problem 2.1. Evaluate the following integral:

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Proof. The answer is

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$$

We use the contour of a semicircle skipping $x = 0$. Let $f(z) = \frac{1 - e^{iz}}{z^2}$, then by Cauchy's integral formula, we have

$$\int_{\gamma_R} f(z) dz + \int_{\gamma_\varepsilon} f(z) dz + \int_{|z| > \varepsilon} f(z) dz = 0 \quad (1)$$

where

$$\lim_{\varepsilon \rightarrow 0} \int_{|z| > \varepsilon} f(z) dz = 2 \int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Hence it suffices to compute the first two terms in (1). Let $z = Re^{i\theta}$, we see that

$$\begin{aligned} \left| \int_{\gamma_R} f(z) dz \right| &= \left| \int_0^\pi \frac{1 - e^{iRe^{i\theta}}}{R^2 e^{2i\theta}} iRe^{i\theta} d\theta \right| \\ &\leq \int_0^\pi \frac{2}{R} d\theta \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Similarly, we can compute $\int_{\gamma_\varepsilon} f(z) dz$ by setting $z = \varepsilon e^{i\theta}$ and letting $\varepsilon \rightarrow 0$.

$$\begin{aligned} \int_{\gamma_\varepsilon} f(z) dz &= \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon^2 e^{2i\theta}} i\varepsilon e^{i\theta} d\theta \\ &= i \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}} d\theta \rightarrow -\pi \end{aligned}$$

as $\varepsilon \rightarrow 0$, since

$$\lim_{a \rightarrow 0} \frac{1 - e^{ia}}{a} = -i$$

Thus the answer is

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$$

□

Problem 2.2. Let U be a region. Suppose f is holomorphic in U , and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n , show that $f \equiv 0$.

Proof. Since $z_n \rightarrow z$, we know $f(z) = \lim_{z_n \rightarrow z} f(z_n) = 0$. Assume f is not identically zero, then writing it as a power series, we have

$$f(w) = (w - z)^n g(w)$$

where $g(w) = a_n + a_{n+1}(w - z) + a_{n+2}(w - z)^2 + \dots$, for some $a_n \neq 0$. We know $g(z) = a_n \neq 0$, hence there exists a neighborhood V around z such that $g(w) \neq 0$ whenever $w \in V$. Hence, if $w \in V$, then $f(w) = 0$ iff $w = z$. This is a contradiction to $z_n \rightarrow z$ and $f(z_n) = 0$ for all n . This shows that a nonzero function has isolated zeros. □

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0 , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$.

Lemma 2.1. Let f be a nonzero holomorphic function, then f cannot have uncountably many zeros.

Proof. Let $Z = \{z \in \mathbb{C} : f(z) = 0\}$, we know that Z contains no limit point. We can tile the complex plane using unit squares and suppose Z is uncountable, the intersection of Z with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. □

□

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k .

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$ because $m > k$. Hence we are done. $\square$

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Proof. (a) We write f as the Taylor series about 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

Using Cauchy's inequality, for any R large, if $n \geq 1$,

$$|f^{(n)}(0)| \leq \frac{n! \sup_{|z|=R} |f(z)|}{R^n} \rightarrow 0$$

as $R \rightarrow \infty$. Thus $f(z) = a_0$, for some $a_0 \in \mathbb{C}$.

(b) The function is not entire. $\square$

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0 , then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. $\square$

Problem 2.7 (Spring 2015, Q6). Suppose f is a nonconstant entire function, then show that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Proof. Suppose there exists $w \in \mathbb{C}$ such that for a small $r > 0$, $D_r(w) \cap f(\mathbb{C}) = \emptyset$. Then we consider the function

$$\frac{1}{f(z) - w}$$

This function is entire and bounded, implying f is constant, which is a contradiction. $\square$

Chapter 3

Singularities

Definition 3.1 (residue). If f has a pole of order n at z_0 , then

$$\operatorname{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z)$$

Proposition 3.1 (Removable singularity). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a removable singularity at z_0 iff

$$\lim_{z \rightarrow z_0} (z - z_0) f(z) = 0$$

iff there exists a neighborhood N of z_0 where f is bounded on N .

Proposition 3.2 (Pole). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a pole at z_0 iff

$$\lim_{z \rightarrow z_0} |f(z)| = \infty$$

iff there exists a non-vanishing holomorphic function h and a unique n such that

$$f(z) = (z - z_0)^{-n} h(z)$$

Proposition 3.3 (Residue theorem). Suppose f is holomorphic in an open set containing a smooth closed curve γ and its interior, except for poles at points $z_1, \dots, z_n$ inside of γ . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^n \operatorname{res}_{z_i} f$$

Proposition 3.4 (Casorati-Weierstrass). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$ and suppose f has an essential singularity at z_0 , then the image of f is dense in $\mathbb{C}$.

Proposition 3.5 (Great Picard's Theorem). If a holomorphic function f has an **essential singularity** at a point w , then on any **punctured neighborhood** of w , $f(z)$ takes on all possible complex values, with at most a single exception, infinitely often.

Proposition 3.6 (Argument principle). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If f has no poles and never vanishes on γ , then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} = (\#\text{zeros of } f \text{ inside } \gamma) - (\#\text{poles of } f \text{ inside } \gamma)$$

where zeros and poles are counted with multiplicities.

Proposition 3.7 (Rouche's theorem). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If

$$|g(z)| < |f(z)| \text{ for all } z \in \gamma$$

then $f, f + g$ have the same number of zeros inside of γ .

Proposition 3.8 (Open mapping theorem). If f is holomorphic and non-constant in a region Ω , then f is open.

Proposition 3.9 (Maximum modulus principle). If f is holomorphic on a region Ω with compact closure $\bar{\Omega}$ and continuous on $\bar{\Omega}$, then

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \partial\Omega} |f(z)|$$

(There are many equivalent statements of MMP, the idea is the same: a nonconstant holomorphic function f cannot attain the maximum in the interior of the compact set).

Proposition 3.10 (Mean-value property). If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta, \quad \text{for any } 0 < r < R.$$

Problem 3.1. Compute

$$\int_0^{\infty} \frac{1}{1+x^2} dx$$

Proof. Let $f(z) = \frac{1}{1+z^2}$. Using a semicircle along the real axis and applying the residue theorem,

$$\int_{|x|<R} \frac{1}{1+x^2} dx + \int_{\gamma_R} f(z) dz = 2\pi i \text{res}_i f$$

where γ_R is the semicircle arc. We first show that $\int_{\gamma_R}$ goes to zero as $R \rightarrow \infty$.

$$\left| \int_{\gamma_R} \frac{1}{1+z^2} dz \right| = \left| \int_0^{2\pi} \frac{1}{1+R^2 e^{2i\theta}} i R e^{i\theta} d\theta \right| = O\left(\frac{1}{R}\right)$$

Then we compute

$$\text{res}_i f = \lim_{z \rightarrow i} (z - i) f(z) = \pi$$

Hence letting $R \rightarrow \infty$, we see

$$\int_0^{\infty} \frac{1}{1+x^2} dx = \frac{\pi}{2}$$

□

Problem 3.2. Compute the following integral: fix $0 < a < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx$$

Proof. Let $f(z) = \frac{e^{az}}{1+e^z}$. We notice that if we use the same semicircle contour then we'd enclose more and more residues, hence now we will use a rectangle contour along the following four lines: real axis, $R + it, t \in [0, 2\pi]$, $-R + it, t \in [2\pi, 0]$, and $s + 2\pi i, s \in [R, -R]$. We notice that

$$\frac{e^{a(x+2\pi i)}}{1+e^{x+2\pi i}} = e^{2a\pi i} \frac{e^{ax}}{1+e^x} = e^{2a\pi i} f(z)$$

thus denoting the two vertical lines as γ_{R+}, γ_{R-} , the residue formula implies

$$(1 - e^{2a\pi i}) \int_{|z| < R} f(z) dz + \int_{\gamma_{R+}} f(z) dz + \int_{\gamma_{R-}} f(z) dz = 2\pi i \operatorname{res}_{\pi i} f$$

We first evaluate

$$\operatorname{res}_{\pi i} f = \lim_{z \rightarrow \pi i} (z - \pi i) f(z) = -e^{a\pi i}$$

Now we show the two vertical pieces tend to 0:

$$\begin{aligned} \left| \int_{\gamma_{R+}} f(z) dz \right| &= \int_0^{2\pi} \left| \frac{e^{a(R+it)}}{1+e^{R+it}} \right| dt \\ &\leq C e^{(a-1)R} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Thus we get

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = \frac{-2\pi i e^{a\pi i}}{1 - e^{2a\pi i}} = \frac{\pi}{\sin \pi a}$$

□

Problem 3.3. Assume that $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if Z is unbounded, then f has an essential singularity at ∞ .

Proof. If Z is unbounded, then we can find a sequence $z_n \in Z$ such that $|z_n| \rightarrow \infty$, which means $f(z_n) = 0$. This shows that ∞ cannot be a pole. Moreover, if ∞ is a removable singularity, then f is bounded for $|z|$ sufficiently large. Using the fact that f is bounded on all compact subsets of $\mathbb{C}$, we see that f is a bounded entire function, thus is identically zero. This is a contradiction. □

Problem 3.4. Suppose f is an injective entire function, show f is linear.

Proof. From the previous problem, we know ∞ cannot be a removable singularity. Moreover, ∞ cannot be an essential singularity because f wouldn't be injective due to Great Picard's theorem. Thus $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$.

Lemma 3.1. Let f be an entire function such that $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, then f must be a polynomial.

(We will prove this in the Spring 2026 exam.) Suppose f has degree ≥ 2 , then for any $w \in \mathbb{C}$, $f(z) - w$ has at least two roots. (For $f(z) - w$ to have repeated roots, $f'(z) = 0$, which only happens at finitely many points because f' is a polynomial). We can choose w such that $f(z) - w$ has two distinct roots, contradicting the fact that f is injective. $\square$

Problem 3.5 (Spring 2017, Q6). Let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function where $U \subset \mathbb{C}$ is an open set containing the closure $\mathbb{D}$ of the unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

If $|f(z)| = 1$ for all $z \in \partial\mathbb{D}$, then prove that

$$\mathbb{D} \subset f(\overline{\mathbb{D}}).$$

Proof. We first show that f must have at least one zero. Suppose not, then $1/f$ is holomorphic in $\mathbb{D}$, and

$$|1/f| \geq 1$$

for $z \in \mathbb{D}$. This is because $|f(z)| = 1$ for $|z| = 1$ implies $|f(z)| \leq 1$ for $z \in \mathbb{D}$. This is a contradiction because $|1/f| = 1$ on $|z| = 1$. Hence f has at least one zero. Now it suffices to show for any $w \in \mathbb{D}$, there exists some $z \in \mathbb{D}$ such that $w = f(z)$. We consider the function $f(z) - w$. It is clear that

$$|-w| < 1 = |f(z)|$$

for $|z| = 1$. Thus $f(z) - w$ and f have the same number of zeros in $\mathbb{D}$, as desired. $\square$

Problem 3.6 (Fall 2019, Q4). Let f be a holomorphic function in the punctured disk

$$\{z : 0 < |z| < 2\}$$

satisfying

$$|f(z)| \leq \left(\log \frac{1}{|z|}\right)^{100} \quad \text{in } \{z : |z| \leq 1/2\},$$

and

$$|f(z)| = 1 \quad \text{on } |z| = 1.$$

(a) Show that f has a removable singularity at the origin.

(b) Show that if $f(z) \neq 0$ in $|z| < 1$, then f is constant.

Proof. (a) We can compute

$$\lim_{z \rightarrow 0} |zf(z)| = 0$$

(b) To show f is constant, one typically should think of Liouville's or MMP (in this case f is not entire, hence it's MMP). We note that $1/f$ defined on $\mathbb{D}$ is a holomorphic function if $f(z) \neq 0$ in $|z| < 1$. Moreover,

$$|f(z)| \geq 1$$

for $|z| \leq 1/2$. However, $|f(z)| = 1$ on $|z| = 1$, which is a contradiction to MMP. $\square$

Chapter 4

Conformal mapping

Definition 4.1 (automorphisms of the disk). Let $a \in \mathbb{D}$, the following function is an automorphism of the disk:

$$\phi_a(z) = \frac{a - z}{1 - \bar{a}z}$$

We have $\phi_a \circ \phi_a = \text{id}$. Moreover, if f is a holomorphic function on $\mathbb{D}$, then $g(0) = 0$ if we let

$$g = \phi_{f(a)} \circ f \circ \phi_a$$

Proposition 4.1 (Schwarz lemma). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$, then

1. $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
2. If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then $f(z) = cz$ for some $|c| = 1$.
3. $|f'(0)| \leq 1$ and if equality holds, then $f(z) = cz$ for some $|c| = 1$.

Proposition 4.2 (Corollary of Schwarz's lemma). If f is an automorphism of $\mathbb{D}$, then there exist $\theta \in \mathbb{R}$ and $a \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{a - z}{1 - \bar{a}z}$$

Namely, the only automorphisms of $\mathbb{D}$ that fix the origin are rotations.

Proposition 4.3 (Automorphisms of $\mathbb{H}$). f is an automorphism of $\mathbb{H}$ iff it is of the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}$, $ad - bc = 1$.

Problem 4.1 (Spring 2019, Q4). Show that the punctured unit disk

$$\{z : 0 < |z| < 1\}$$

and the annulus

$$\{z : 1 < |z| < 2\}$$

cannot be conformally equivalent.

Proof. Suppose $f : \mathbb{D} \setminus \{0\} \rightarrow \mathcal{A}$ is a conformal map, where $\mathcal{A}$ denotes the annulus. Intuitively, we will show how the punctured point 0 is different from a punctured disk. Since $|f(z)| < 2$, we know f has a removable singularity at $z = 0$, where we denote its holomorphic extension as $\tilde{f}$. We first note by continuity $1 \leq |\tilde{f}(z)| \leq 2$.

1. We claim $\tilde{f}(0) \neq 2$ because that'd imply an interior point attained the maximum, contradiction to MMP.
2. Similarly, $\tilde{f}(0) \neq 1$ as we consider $1/\tilde{f}$, which is holomorphic on $\mathbb{D}$ and $|1/\tilde{f}(z)| \leq 1$. It would again imply $1/\tilde{f}$ attained maximum at an interior point.

Thus $1 < |\tilde{f}(z)| < 2$. Now there must exist $z_0 \in \mathbb{D}$ such that

$$\tilde{f}(z_0) = \tilde{f}(0)$$

Now we choose $\varepsilon > 0$ small enough, such that $D_\varepsilon(0) \cap D_\varepsilon(z_0) = \emptyset$. By the Open Mapping Theorem, we know

$$f(D_\varepsilon(0)), f(D_\varepsilon(z_0)) \text{ are open sets containing } f(z_0) = \tilde{f}(0)$$

Letting $D_\delta(f(z_0))$ denote an open disk contained in both of the open sets, then

$$f^{-1}(D_\delta(f(z_0))) \subset D_\varepsilon(0), f^{-1}(D_\delta(f(z_0))) \subset D_\varepsilon(z_0)$$

which is a contradiction. □

Problem 4.2 (Fall 2019, Q4). Let $f(z)$ be a holomorphic function on

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\},$$

with $|f(z)| < 1$, and suppose $f(\alpha) = 0$ for some $|\alpha| < 1$. Show that for $z \in \mathbb{D}$,

$$|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right|.$$

Proof. We consider the following composite function:

$$g = f \circ \phi_\alpha(z)$$

It is clear that $g(0) = 0$ and $|g(z)| < 1$. Applying Schwarz lemma, we get

$$|f \circ \phi_\alpha(z)| \leq |z|$$

Since ϕ_α is an automorphism of $\mathbb{D}$, replacing z with $\phi_\alpha(z)$ gives

$$|f(z)| \leq |\phi_\alpha(z)|$$

as desired. □

Problem 4.3 (Spring 2016, Q5). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic function where

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

is the unit disk. Prove that if f has at least 2 fixed points then f is the identity map.

Proof. Let $z_1, z_2 \in \mathbb{D}$ be such that $f(z_1) = z_1, f(z_2) = z_2$. Then we consider the composite

$$g = \phi_{z_1} \circ f \circ \phi_{z_1}$$

It is clear that g fixes 0, and if we let $z'_2 \neq 0$ be such that $\phi_{z_1}(z'_2) = z_2$, then it is clear that g fixes z'_2 as well. We see that

$$g(0) = 0, \quad |g(z)| \leq 1, g(z'_2) = z'_2$$

hence applying the Schwarz lemma, there exists $|c| = 1$ such that $g(z) = cz$, which implies g is constant. This shows f is constant as well. $\square$

Chapter 5

Spring 2026

Problem 5.1. Let f be a holomorphic function on $\{z : |z| < 1\}$ that extends to a continuous function on $\{z : |z| \leq 1\}$ and maps the closed unit disk to itself. Furthermore, suppose

$$f(0) = f'(0) = \cdots = f^{(n-1)}(0) = 0.$$

Show that:

(1)

$$|f(z)| \leq |z|^n \quad \text{for all } |z| \leq 1.$$

(2) If there is some $P \in D(0, 1) \setminus \{0\}$ such that

$$|f(P)| = |P|^n$$

or if

$$|f^{(n)}(0)| = n!,$$

then

$$f(z) = e^{i\theta} z^n.$$

Proof. (1) If $g(z) = f(z)/z^n$ is well-defined on $\bar{\mathbb{D}}$, then it suffices to show

$$|g(z)| \leq 1$$

We note that on $|z| = 1$,

$$|g(z)| = \left| \frac{f(z)}{z^n} \right| = |f(z)| \leq 1$$

because $f : \bar{\mathbb{D}} \rightarrow \bar{\mathbb{D}}$. Now it suffices to define g carefully:

$$g(z) = \begin{cases} f(z)/z^n, & z \neq 0 \\ f^{(n)}(0)/n!, & z = 0 \end{cases}$$

It is clear that g is holomorphic on $\bar{\mathbb{D}}$.

(2) If $|f(P)| = |P|^n$ for some $P \neq 0, |P| < 1$, then $|g(P)| = 1$ at an interior point. Since $|g| \leq 1$, by the Maximum Modulus Principle, g must be a constant, i.e., there exists some θ such that

$$g(z) = e^{i\theta}$$

as desired. The other case is exactly the same. □

Problem 5.2 (Also Fall 2023, Q2). Let $-1 < a < 0$. Compute

$$\int_0^\infty \frac{x^a}{1+x} dx.$$

Proof. We computed earlier that for $0 < b < 1$,

$$\int_{-\infty}^\infty \frac{e^{by}}{1+e^y} dy = \frac{\pi}{\sin \pi b}$$

Thus letting $x = e^y$, we have

$$\int_0^\infty \frac{x^a}{1+x} dx = \int_{-\infty}^\infty \frac{e^{(a+1)y}}{1+e^y} dy = \frac{\pi}{\sin \pi(a+1)}$$

Alternatively, one can use a keyhole contour near zero and the residue formula for the residue at $z = -1$. $\square$

Problem 5.3. Two parts.

(1) Let

$$f : U = \{z : 0 < |z| < 1\} \longrightarrow \mathbb{C}$$

be holomorphic and nonconstant. If there exists a sequence $\{z_n\} \subset U$ that converges to 0 and satisfies

$$f(z_n) = 0 \text{ for all } n$$

then prove that 0 is an essential singularity of f .

(2) Show that if f is an entire function that satisfies

$$\lim_{|z| \rightarrow \infty} |f(z)| = \infty,$$

then f is a polynomial.

Proof. (1) It is clear that 0 is not a pole. If 0 were to be a removable singularity for f , then we can have a holomorphic extension F defined on $\mathbb{D}$. However, we've shown earlier that the zeros of holomorphic functions must be isolated, which is a contradiction to f be nonconstant. This shows that 0 is an essential singularity.

(2) We know that ∞ is a pole. Now consider the Laurent series

$$g(z) = f\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} a_n z^{-n}$$

by writing f as $\sum_n a_n z^n$, and 0 is a pole for g . This implies g must have finitely many nonzero coefficients, as desired. $\square$

Problem 5.4. Describe all injective holomorphic maps that map

$$\mathbb{C} \cup \{\infty\} \text{ to } \mathbb{C} \cup \{\infty\}$$

and also map

$$\mathbb{R} \cup \{\infty\} \text{ to } \mathbb{R} \cup \{\infty\}.$$

Proof. Let f be such a function, then by the Open Mapping Theorem, f is open. Now viewing $\mathbb{C} \cup \{\infty\}$ as the (compact) Riemann sphere, $f(\mathbb{C})$ is both open and closed, thus must be an automorphism. Since f also fixes $\mathbb{R} \cup \{\infty\}$, let $\mathbb{H}$ denote the upper half plane, then either $\mathbb{H} \mapsto \mathbb{H}$ or $\mathbb{H} \mapsto -\mathbb{H}$. In the first case, we know f must take the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}, ad - bc = 1$. In the second case, f can be written as the composite of the function $z \mapsto -z$ and a mobius transformation. Combined, all the maps are of the form

$$\left\{ \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{R}, ad - bc = \pm 1 \right\}$$

□